Grid Pulse EV · Project J26-DS-331 · Component 3

Optimizing Public EV Charging Schedules for Renewable-Curtailment Absorption and Peak Reduction in Sri Lanka

A complete walkthrough of the research: the plan, every step taken, why each method was chosen, the alternatives, the earlier research it follows, the models trained, how results are validated, and every change made on 19 October 2026

Student	Wijesuriya W. A. A. I. (IT23274648)
Supervisor	Prof. Samantha Rajapaksha
Co-supervisor	Ms. Thamali Dassanayake
Programme	B.Sc. (Hons) Information Technology, specialising in Data Science, SLIIT
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Status of the results in this document

Every number comes from the project's own result files and can be regenerated. The minimum secure level (NLmin = 520 MW), the inertia (SNSP) limit and the scenario years are now **cited to CEB's Long Term Generation Expansion Plan 2025-2044**, but still need review by a CEB System Control engineer. The EV session data and the fleet size are still **provisional placeholders**. The results show how the method behaves; they are **not yet dissertation findings**.

Several method changes from the approved proposal (listed in Section 8) still need supervisor approval.

Contents

0. About this document 4

0.1 Who did what . . . 4

0.2 How to read it . . . 4

1. The research problem 5

1.1 The problem in plain words . . . 5

1.2 Evidence that the problem is real . . . 5

1.3 The research gap . . . 5

1.4 Research question and objectives . . . 5

2. Research design 6

2.1 Type of research . . . 6

2.2 The controlled numerical experiment . . . 6

2.3 Why optimisation (linear programming) . . . 6

2.4 The pipeline . . . 6

2.5 Key quantities (read Appendix A for the full glossary) . . . 6

3. The research task plan 7

4. Step by step: how the research was done 8

4.1 Data acquisition . . . 8

4.2 Preprocessing and quality audit . . . 8

4.3 Exploratory grid analyses . . . 9

4.4 Hourly net-demand series NL(t) . . . 9

4.5 Reconciliation with CEB published totals . . . 10

4.6 Review of the work against the proposal and the literature . . . 10

4.7 System-boundary correction and available solar (SO1) . . . 11

4.8 The curtailment-risk signal C(t) (SO2) . . . 12

4.8a Deriving NLmin from CEB's own planning values (Appendix C) . . . 13

4.8b Scenario years from the CEB expansion plan . . . 14

4.9 Charging sessions (SO3) . . . 15

4.10 Charging strategies and the LP scheduler (SO4) . . . 15

4.11 The forecasting layer: the models that were trained (WBS 5.8) . . . 17

4.12 Experiment design (SO5 and SO6) . . . 17

5. Results (provisional fleet and session inputs) 18

5.1 Absorption on the curtailment-risk days (SO5) . . . 18

5.2 The year-representative sample (SO3, SO5) . . . 21

5.3 Forecast accuracy (models trained) . . . 22

5.4 Trade-off between absorption, peak and delay (SO5) . . . 22

5.5 Sensitivity: is the ranking preserved? (SO6) . . . 23
5.6 Scalability and speed (NFR1, NFR2) . . . 24
6. Which models exist, and which were trained 25
7. How every result is validated 26
8. Every change made on 19 October 2026 26
8.1 Corrections to existing work . . . 26
8.2 New components . . . 26
8.2a Second round of changes (same day, after review of the first results) . . . 27
8.3 Deviations from the approved proposal (need supervisor approval) . . . 27
8.4 Problems found in the proposal document itself . . . 28
9. Earlier research followed, and what was taken from each 28
10. What is new in this research (novelty) 29
10.1 Not new - cite these as foundations . . . 29
10.2 New in this component . . . 29
10.3 Position against CEB's own EV study . . . 29
11. Limitations and threats to validity 30
12. What to do next 30
13. Progress Presentation 1: evidence for each marking criterion 31
13.1 Requirements traceability (proposal Tables 5.1 and 5.2) . . . 32
13.2 Risk register . . . 32
13.3 Suggested slide outline (about 12 slides) . . . 33
Appendix A. Glossary 34
Appendix B. How to run everything 34
Appendix C. Parameters (config/scenarios.json) 35
Appendix D. Likely viva questions, with answers 35

0. About this document

This document explains the whole research from the beginning, so that you can understand, explain and defend every part of it. Each stage of the work is described in the same order: **what** was done, **how**, **why** that way, which **alternatives** existed, which **earlier research** it follows, how it was **validated**, and the **result**.

0.1 Who did what

Your examiners will ask about every part, and your AI-use disclosure must describe the help you received. This table separates the work recorded in your repository before 19 October from the work added in the AI-assisted session (Claude Code) on 19 October under your direction.

Work	Source	Date
Topic, research gap, objectives SO1-SO6, proposal	Your proposal	Aug-Sep 2026
PUCSL preprocessing, quality flags, monthly audits, consolidation, tests	Your repository history	13-15 Oct
Curtailment-pressure and coal-commitment analyses, hourly NL(t)	Your repository history	15 Oct
CEB reconciliation and January 2024 source-level audit	Your repository history	15 Oct
NASA POWER 2025 pilot; EV prototype (140 sessions)	Your repository history	14-19 Oct
Review of repo against proposal and literature; corrections; grid model; session model; LP scheduler; forecasting models; experiments; methods document	AI-assisted session, directed by you	19 Oct
NLmin derived from the CEB LTGEP; LTGEP scenario years; risk-day evaluation; parallel solving; full sweeps; this document updated	AI-assisted session, directed by you	19 Oct
Speed benchmark; interactive dashboard (WBS 5.11)	AI-assisted session, directed by you	19-20 Oct

> **What you must be able to do**
>
> For the work added on 19 October you must understand the code well enough to explain it without help, and you must declare the assistance in your AI-use disclosure. Sections 4.6 to 4.12 and Appendix D (viva questions) are written for that purpose.

0.2 How to read it

- Sections 1-3: the problem, the research design and the task plan.
- Section 4: every stage of the work in order, from data to experiments. This is the core.
- Sections 5-7: results so far, the models that were trained, and how everything is validated.
- Sections 8-12: all changes made, the literature followed, novelty, limitations and next steps.
- Section 13: the work mapped to the Progress Presentation 1 mark sheet, with requirements traceability, a risk register and a slide outline.
- Appendices: glossary, how to run the code, the parameter file, and likely viva questions.

1. The research problem

1.1 The problem in plain words

Sri Lanka's electricity system now has a lot of solar power at midday but not much demand at that time, especially on weekends and holidays. Coal and other large spinning (synchronous) generators cannot be turned down below a minimum safe level, because they keep the grid stable. When solar plus this minimum is more than demand, the operator must switch some solar off. This is called **curtailment**: clean energy that was available is thrown away.

In the evening the opposite happens: solar falls to zero, demand peaks, and expensive coal and oil plants run. Public electric-vehicle (EV) charging currently happens mostly on arrival, in the late afternoon and evening, so it adds load at the worst time.

1.2 Evidence that the problem is real

- In 2025 CEB asked rooftop-solar owners to switch off during the April New Year holiday and ordered large rooftop systems off; solar farms were shut down at midday on weekends and holidays during 2025, and reportedly on some weekdays in early 2026.
- The 9 February 2025 island-wide blackout was publicly linked to low system inertia at high solar output on a low-demand day.
- Your own data analysis (Section 4.3) shows the midday dip deepening: in 1-18 March, the share of days with midday dispatch below night-time dispatch rose from 16.7% in 2024 to 61.1% in 2025.
- In March 2026 the President asked EV owners to charge in daytime, saying evening charging added about 300 MW (cited in your proposal).

1.3 The research gap

Curtailment-aware EV charging has been studied, but only where curtailment is **measured and published** (for example Great Britain, Dixon et al. 2020). Sri Lanka publishes no hourly curtailment record, and here curtailment is caused mainly by minimum-generation and inertia limits rather than by transmission congestion. Nobody has scheduled public EV charging against an **inferred** curtailment signal in such an island system and measured the cost to drivers.

1.4 Research question and objectives

> **Research question**
>
> How much renewable energy at risk of curtailment can be absorbed by rescheduling public EV charging in Sri Lanka, how much does the evening peak fall, and what does it cost drivers in delay, when no measured curtailment data exists?

No.	Specific objective (from the proposal)	Measurable outcome	Status (19 Oct)
SO1	Acquire, validate and time-align 12+ months of demand and generation by source; build NL(t)	Validated hourly NL(t) reconciled to CEB published figures	Done for 2024; reconciled within 0.1%
SO2	Derive NLmin; infer hourly curtailment risk C(t); identify peak windows	C(t) series, documented NLmin derivation, cross-check with observed minimum	Done: NLmin 520 MW from LTGEP values; 0 observed hours below it; CEB review pending
SO3	Quantify uncontrolled charging impact with realistic sessions	Added evening peak (MW), unabsorbed risk energy (MWh) per fleet scenario	Done with placeholder sessions; ACN-Data fit pending
SO4	Curtailment-aware LP scheduler with session and station constraints	Feasible schedules, 100% completion, zero violations	Done; zero violations

No.	Specific objective (from the proposal)	Measurable outcome	Status (19 Oct)
SO5	Compare with uncontrolled and solar-following using grid and service metrics	Metrics table and weight trade-off curve	Done (weekly sample + every risk day)
SO6	Robustness to NLmin and to share of drivers declaring departure	Sensitivity results; is the strategy ranking preserved?	Done (42 days, 16 settings)

2. Research design

2.1 Type of research

This is **quantitative, simulation-based experimental research** using mathematical optimisation. The question is **counterfactual**: what would happen if charging were scheduled differently? That schedule has never been operated, so it cannot be observed; it must be simulated on real grid conditions.

2.2 The controlled numerical experiment

For every comparison, the **same charging sessions** and the **same grid hours** are used for every strategy. The only thing that changes is the charging strategy. Any difference in the results is therefore caused by the strategy. Dixon et al. (2020), the closest earlier study, uses the same design, so your results are comparable in form.

2.3 Why optimisation (linear programming)

- The physics is linear: energy = power × time, and grid balance is a sum of powers.
- A linear program (LP) is solved to the proven best answer (global optimum), so the reported benefit is the best achievable under the constraints, not the output of a heuristic.
- It needs no training data, which matters because no Sri Lankan charging data exists.
- It is fully reproducible: same inputs, same answer.

2.4 The pipeline

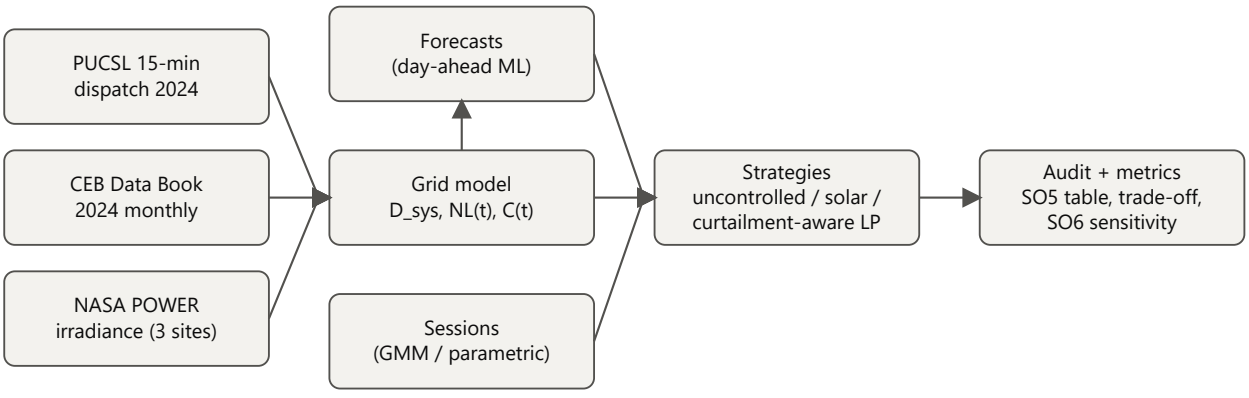


Figure 1. The research pipeline. Every box is a separate, tested script; every arrow is a file on disk, so each stage can be checked on its own.

2.5 Key quantities (read Appendix A for the full glossary)

Symbol	Meaning	Unit
D(t)	Total dispatch reported by PUCSL (transmission level)	MW
$D_{sys}(t)$	System demand after adding back embedded generation that PUCSL leaves out	MW

Symbol	Meaning	Unit
NL(t)	Residual (net) load: D_{sys} − solar − wind − must-take. The load coal, oil and major hydro must serve.	MW
NLmin	Minimum secure level of synchronous generation	MW
SNSP	System non-synchronous penetration: the maximum share of demand that inverter-based sources (solar, wind) may supply at once	%
C(t)	Inferred curtailment risk: how much renewable power would have to be curtailed	MW
L(t)	Public EV charging load	MW

3. The research task plan

The work breakdown structure (WBS) from your proposal, with the current status. You are about two months ahead of the plan: most remaining work is replacing provisional values with real ones, not writing new code.

WBS	Task	Planned	Status on 19 Oct 2026
5.1	Acquire and preprocess demand, generation and solar data; build NL(t)	Oct-Dec 2026	Done for 2024; 2025 data recommended
5.2	Derive NLmin; document and cross-check	Nov-Dec 2026	Done from LTGEP; CEB review pending
5.3	Compute C(t); identify peak-stress windows	Dec 2026-Jan 2027	Done; LTGEP scenario years 2025-2030
5.4	Fit session distributions; session generator	Dec 2026-Jan 2027	Done; waiting for ACN-Data
5.5	Uncontrolled charging baseline	Jan 2027	Done
5.6	LP scheduler	Jan-Feb 2027	Done
5.7	Solar-following baseline and comparison harness	Feb 2027	Done
5.8	Forecasting layer, benchmarked against persistence	Feb-Mar 2027	Done (no weather-forecast inputs yet)
5.9	Weight sweeps; trade-off curve	Mar 2027	Done (on every risk day)
5.10	NLmin sweep and scenario sensitivity	Mar 2027	Done (42 days)
5.11	Dashboard views	Mar-Apr 2027	Done early (20 Oct): live scheduling dashboard
5.12	Expert review (CEB System Control)	Apr 2027	Not started
5.13	Final analysis, dissertation, paper	Apr-Jun 2027	Not started

4. Step by step: how the research was done

4.1 Data acquisition

What

Collected 15-minute generation dispatch reports published by the Public Utilities Commission of Sri Lanka (PUCSL) for January-December 2024 and 1 January-18 March 2025: one Excel workbook per month, one sheet per day, eight generation sources plus a total.

How it was done

- Downloaded the monthly workbooks from the PUCSL website into `data/raw/grid/pucsl/`.
- Kept the original files untouched; all processing works on copies.
- Recorded each dataset in `docs/data_inventory.csv` (source, period, resolution, limitations).

Why this way

Your proposal planned on CEB hourly demand data, but CEB does not publish an hourly archive. PUCSL does publish 15-minute data, which is finer than the hourly resolution SO1 requires. 2024 gives a complete calendar year.

Alternatives that existed, and why they were not chosen

Alternative	Why not chosen
Request SCADA data from CEB	Possible later, but slow and uncertain; not available in time
Use CEB daily summaries	Too coarse to see the midday dip
Use synthetic grid data	Not evidence; only acceptable for tests

Earlier research or standard practice followed

Using official published system data and reconciling it to published totals is the standard approach when operator data is not shared.

How it was validated

Coverage: 35,040 of 35,136 quarter-hours in 2024 observed (99.73%); one day (30 June 2024) missing in the source.

4.2 Preprocessing and quality audit

What

Converted every daily sheet into one clean, machine-readable table with exact timestamps and a quality flag on each 15-minute interval.

How it was done

- `preprocess_pucsl_dispatch.py` reads each sheet, checks the date stamp and the time headers, and extracts the eight sources and the reported total.
- Time labels 00:15 ... 00:00 are treated as the *end* of each 15-minute interval, so the interval 00:00-00:15 is labelled 00:15.
- Problems are **flagged, not fixed**: negative values, totals that differ from the sum of sources by more than 5 MW, wrong date stamps, wrong time headers.
- Months are consolidated into one file per year; an audit document records each period's exceptions; automated tests cover each month's quirks.

Why this way

The project rule is to never invent data. Interpolating or imputing would create values that look measured but are not. Flagging keeps every problem visible and lets later analyses decide what to exclude.

Alternatives that existed, and why they were not chosen

Alternative	Why not chosen
Interpolate missing or bad values	Creates data that looks measured; hides problems
Delete bad days silently	Loses information and cannot be audited
Manual cleaning in Excel	Not reproducible; cannot be re-run when new months arrive

Earlier research or standard practice followed

Open power-system data practice (flag, document, never silently impute); reproducible research practice (scripted pipeline, version control, tests).

How it was validated

- 1,263 intervals in 2024 flagged for review; January-March 2025: 7,392 intervals, 54 flagged.
- Month-specific edge-case tests; the test suite now has 159 passing tests.

4.3 Exploratory grid analyses

What

Two exploratory analyses to understand whether the midday problem is real and how close coal is to its minimum.

How it was done

- **Curtailment pressure** (`analyse_curtailment_pressure.py`): counts days on which the midday minimum of dispatch fell below the night-time minimum (the 'duck curve' sign), by month and year.
- **Coal commitment** (`analyse_coal_commitment.py`): estimates how many coal units ran (output steps of about 273 MW = 99th percentile ÷ 3 units) and how much room was left above an empirical per-unit floor (154-156 MW) in the lowest 2% of midday intervals.

Why this way

Before building a model, you needed evidence that midday surplus exists and is growing, and a rough idea of how flexible coal is.

Earlier research or standard practice followed

The duck-curve indicator is widely used (CAISO, NREL). The coal heuristics are your own and have no direct precedent, so they must stay exploratory.

How it was validated

- You corrected your own first estimate: ~160 MW of headroom became 60-74 MW after you found you had compared two-unit operation with a one-unit floor. Recording this in the log is good scientific practice.
- Both scripts now exclude flagged intervals and require complete days.

Result

- February: 17% of days in 2024 vs 52% in 2025; 1-18 March: 16.7% vs 61.1%. Mean midday minimum fell from 1,731.7 MW to 1,390.4 MW.
- 166 stressed intervals; mean coal headroom 74 / 60 / 19 / 11 MW for 2 / 4 / 8 / 24-hour commitment windows; coal within 10% of its floor in 27-61% of them. These are sensitivity estimates, not physical limits.

4.4 Hourly net-demand series NL(t)

What

Turned the 15-minute data into an hourly series, because SO1 asks for hourly or half-hourly data and the scheduler works on hourly grid conditions.

How it was done

- `build_hourly_net_demand.py` averages the four quarter-hours that start in each local hour (mean MW over one hour = MWh in that hour).
- An hour is published only if all four quarter-hours exist and none is flagged; otherwise it is left empty with a status (missing, incomplete, contains_flag).

Why this way

Averaging power is the correct way to change resolution (energy is conserved). Leaving bad hours empty keeps the 'never impute' rule.

How it was validated

Hourly energy equals the daily integrated energy exactly on all 255 fully complete days.

Result

8,784 hours in 2024: 8,326 complete, 434 containing a flag, 24 missing (30 June). June is the weakest month (153 unusable hours).

4.5 Reconciliation with CEB published totals

What

Compared PUCSL energy with the official CEB *Sales and Generation Data Book 2024* (pages 73-74), month by month and source by source.

How it was done

- Transcribed CEB's monthly gross and net generation by source into `docs/ceb_2024_monthly_reference.csv`; checked the monthly sums against the published annual totals.
- `reconcile_ceb_2024.py` integrates PUCSL 15-minute power to energy (MW × 0.25 h) and compares it with the PUCSL daily totals and the CEB figures.

Why this way

SO1 requires the series to be 'reconciled against published CEB aggregate figures'. Without this you cannot say what PUCSL dispatch actually measures.

How it was validated

Coal from PUCSL matches CEB net coal within 11.8 GWh (0.2%), which shows the 15-minute values are sent-out (net) average power.

Result

- PUCSL 15-minute total: 15,402 GWh, about 8% below CEB net generation (16,802 GWh).
- The gap between PUCSL daily totals and the 15-minute data is 988 GWh, of which 816 GWh (83%) is solar - close to CEB's rooftop-solar energy (867 GWh).
- **Key discovery:** the 15-minute series leaves out rooftop solar and other small embedded generators. PUCSL utility solar 203 GWh vs CEB 257 GWh; wind 717 vs 797 GWh; mini hydro 1,127 vs 1,473 GWh. Everything in Section 4.7 is built on this finding.

4.6 Review of the work against the proposal and the literature

On 19 October the repository was reviewed in three ways: (1) every number in the research log was re-computed from the data; (2) the work was compared with the approved proposal; (3) each method was compared with published practice. All headline numbers reproduced exactly. The review found these problems:

#	Problem found	Why it matters	Fixed by
1	Curtailment risk was going to be computed from **metered** solar	Metered output is recorded after curtailment, so C(t) would be close to zero	Section 4.7
2	Only 2024 data	Real curtailment started in 2025; 2024 has little to find	Recommended (decision needed)
3	Synthetic sessions uniform and uncalibrated; 19 of 140 could never be fully charged	Conflicts with the 100% completion constraint; results not realistic	Section 4.9
4	Mini hydro treated as dispatchable	It is run-of-river; worth ~90-120 MW at midday	Section 4.7 (must-take)
5	Proposal's NLmin formula adds reserve and 'inertia MW'	Double-counts units; inertia is not MW	Section 4.8
6	Coal and pressure analyses are home-made heuristics	No precedent	Kept as exploratory only
7	Research log missing the CEB reconciliation; NLmin worksheet out of date; data inventory missing PUCSL; README stale	Documentation must match the evidence	Section 8.4
8	Proposal document: citation numbers wrong, appendices mismatched, AI disclosure copied from another project	Examiners will notice	You must fix (Section 12)

4.7 System-boundary correction and available solar (SO1)

What

Built `build_grid_scenario.py`, which turns the PUCSL data into true system demand and models the solar that was *available*, not the solar that was dispatched.

How it was done

- **Boundary correction.** Each source is rescaled so its monthly energy equals CEB's: solar = NASA irradiance shape × CEB monthly (grid + rooftop) solar; wind = metered wind × CEB/PUCSL monthly ratio; must-take (mini hydro + biomass) = metered × monthly ratio. The missing embedded generation is added back to demand:
- $D_{sys}(t) = D(t) + (S - solar_{metered}) + (W - wind_{metered}) + (MT - MT_{metered})$
- **Available solar.** NASA POWER hourly irradiance was downloaded for Colombo (where most rooftop solar is), Hambantota (south) and Jaffna (north). The three-site mean gives the hour-by-hour shape; CEB's monthly energy gives the size.
- **Time alignment.** NASA reports UTC hours, which fall at HH:30 in Sri Lanka. Each local hour is the mean of the two NASA hours that overlap it. The code refuses to run if there is sunlight at night (which would mean a time-zone error).
- **Must-take.** Mini hydro and biomass are subtracted as must-take, not counted as dispatchable.

Why this way

- Metered solar hides curtailment: if solar was switched off, the meter shows less solar. Every curtailment study models available renewable output instead.
- Scaling to official monthly energy uses only published numbers, so no unverified installed-capacity figures are needed.

Alternatives that existed, and why they were not chosen

Alternative	Why not chosen
Use metered solar	Hides curtailment; C(t) ≈ 0

Alternative	Why not chosen
Full PV model (e.g. pvlib) with installed capacity per site	Needs capacity and location data you cannot yet cite
Commercial satellite PV data (e.g. Solcast)	Paid; not open; not reproducible by examiners

Earlier research or standard practice followed

Denholm & Margolis (2007) simulate hourly PV output from irradiance and compare it with what the system can absorb, for exactly this reason.

How it was validated

- Solar shape correlation with PUCSL metered solar: 0.963 (Colombo 0.938, Hambantota 0.965, Jaffna 0.948).
- Modelled solar at night: 0.0 MW.
- Corrected demand, scaled to a full year: 16,786 GWh vs CEB net generation 16,802 GWh (−0.1%). **This is the SO1 reconciliation the proposal asks for.**
- Identity check: with today's solar, the model reproduces observed coal + oil + major hydro output hour by hour, so the 2024 baseline is observed data, not a model.

4.8 The curtailment-risk signal C(t) (SO2)

What

Defined the inferred curtailment risk from the two constraint types island operators use.

How it was done

- Residual load: $NL(t) = k_D \cdot D_{sys}(t) - m_{solar} \cdot S(t) - m_{wind} \cdot W(t) - k_{MT} \cdot MT(t)$. The multipliers come from the scenario year (Section 4.8b); all are 1 for 2024.
- Floor constraint: synchronous plants cannot go below NLmin, so any extra renewable power must be curtailed: $C_{floor}(t) = \max(0, NLmin - NL(t))$.
- Inertia constraint: inverter-based power may supply at most a share SNSP of demand: $C_{snsp}(t) = \max(0, VRE(t) - SNSP \cdot D_{sys}(t))$.
- $C(t) = \max(C_{floor}, C_{snsp})$. Peak-stress windows are hours within 3% of each day's maximum NL.

Why this way

- The proposal's formula adds minimum generation + spinning reserve + an 'inertia MW' term. That double-counts: the same running units provide all three, and inertia is measured in MW·s, not MW. The floor-plus-SNSP form avoids both errors, and it is the form CEB itself uses in its plan (an SNSP limit).

Alternatives that existed, and why they were not chosen

Alternative	Why not chosen
Full unit-commitment or production-cost model (e.g. PLEXOS)	Needs detailed unit data for every plant; a project in itself
NLmin = a percentile of observed data	Circular: assumes the answer
Additive formula from the proposal	Double-counts; dimensionally wrong for inertia

Earlier research or standard practice followed

- Denholm & Margolis (2007): renewable output becomes unusable when the inflexible conventional fleet cannot turn down further - the 'minimum generation' approach.
- EirGrid (Ireland) operates its island grid with an SNSP limit and a minimum number of synchronous units; CEB's LTGEP 2025-2044 plans with an SNSP limit of 75%.

How it was validated

Signal plausibility (proposal §3.5), central scenario: 89% of risk energy falls between 10:00 and 15:00, and risk is 250 MWh per weekend day vs 27 MWh per weekday. This matches the real 2025 pattern of weekend and holiday curtailment.

4.8a Deriving NLmin from CEB's own planning values (Appendix C)

What

Replaced the 700 MW placeholder with a derived minimum secure level, every plant and policy value cited to the CEB *Long Term Generation Expansion Plan 2025-2044* (LTGEP).

How it was done

- **Cited values.** Lakvijaya coal: 3 units of 270 MW, minimum operating level **150 MW** per unit (Table 2.7, p. 2-10). Spinning reserve 5% of demand and no unit above 25% of demand (p. 7-13). SNSP limit 75% (pp. 7-7, 7-13, 10-14). Mini hydro is 'non-dispatchable run-of-river' (p. 2-3).
- **Observed practice.** In the 167 most stressed hours of 2024 (lowest 2% of coal + oil + major hydro output), the number of coal units online was 1 in 4 hours, 2 in 136 and 3 in 27. Major hydro kept online: minimum 90.5 MW, 10th percentile 217.0 MW, median 329.0 MW. Oil was off in 77% of them.
- **Formula:** NLmin = coal units online × 150 MW + major hydro kept online.
- **Reserve and the 25% rule are checks**, not addends: reserve is upward headroom on the units already running.

Why this way

- Full unit data (inertia constants, ramp rates) is not public, so a full unit-commitment model cannot be cited. Using the cited plant minimum together with the commitment the operator actually chose is the most defensible floor available from public sources.
- Fixing the commitment from the most stressed hours means the floor is what the operator did when the system was closest to its limit, not an average day.

Alternatives that existed, and why they were not chosen

Alternative	Why not chosen
Keep 700 MW placeholder	Uncited; observed operation went below it in 32 hours of 2024
Lowest observed dispatchable output (575.5 MW)	Circular: assumes 2024 was at its limit
Full unit commitment with H and S per unit	Unit data not published; ask CEB for it

Earlier research or standard practice followed

Minimum-generation curtailment (Denholm & Margolis 2007); island operation with a minimum number of synchronous units and an SNSP limit (EirGrid; CEB LTGEP).

How it was validated

- Reserve: the largest requirement in the stressed hours is 89.8 MW; two units at minimum leave 240 MW of headroom, so it is met in 100% of them. The 25% single-unit rule holds in 100%.
- **Cross-check (SO2):** observed 2024 dispatchable output fell below the central floor in **0 hours** (observed minimum 575.5 MW). The old 700 MW placeholder was crossed in 32 hours, so it was too high.
- SNSP: the highest VRE share of demand in 2024 was 46.3%, well below 75%.
- Remaining check: review by a CEB System Control engineer (proposal §3.5).

Scenario	Coal units × 150 MW	Major hydro kept online	NLmin
Low	1 × 150	minimum (90.5 MW)	240 MW
Central	**2 × 150**	**10th percentile (217.0 MW)**	**517 MW → 520 MW**

Scenario	Coal units × 150 MW	Major hydro kept online	NLmin
High	3 × 150	median (329.0 MW)	779 MW

The sensitivity sweep uses 240, 380, 520, 650, 780 MW: the three scenarios and the two midpoints. The high value (about 780 MW) represents the more cautious operation seen after the 9 February 2025 blackout.

4.8b Scenario years from the CEB expansion plan

What

Replaced the uncited 'solar × 1.5' with scenario years 2025-2030 taken from the LTGEP base case, so that solar, wind, must-take generation and demand all grow consistently.

How it was done

- **Installed at end of 2023** (LTGEP Table 2.1, p. 2-1): solar 953 MW (815 MW of it rooftop), wind 267 MW; mini hydro 419 + biomass 54 MW (Table 2.3).
- **Planned additions per year** (Table E3, p. E-7), e.g. solar +370 MW in 2026, +400 MW in 2027, +450 MW per year in 2028-2030. 2024 additions are not in the plan and are set to the plan's 2025 rate (200 MW); 0-400 MW changes the multipliers by less than 0.15.
- **Multiplier** = mid-year installed capacity in year Y ÷ mid-2024 capacity. Demand multiplier = LTGEP net-generation forecast (Table 3.3, p. 3-11) ÷ CEB 2024 actual (16,802 GWh).
- Each 2024 hourly profile keeps its shape and is scaled; the 2024 weather is reused.

Why this way

- Scaling solar alone ignores demand growth. For 2028 that would give 133 GWh of risk instead of 31 GWh, four times too much.
- Using the operator's own plan makes the scenario defensible in front of CEB.

Alternatives that existed, and why they were not chosen

Alternative	Why not chosen
Single solar multiplier	Uncited; ignores demand and wind growth
Forecast own capacity growth	Would compete with the official plan without better data

Earlier research or standard practice followed

Scaling a reference-year hourly profile by planned capacity and energy is how expansion plans (including the LTGEP) build future hourly cases.

How it was validated

The 2024 row reproduces observed operation exactly (all multipliers 1, zero risk at NLmin 520 MW).

Year	Solar ×	Wind ×	Demand ×	Risk hours	Risk energy	Days with risk
2024	1.00	1.00	1.00	0	0.0 GWh	0
2025	1.19	1.02	1.05	0	0.0 GWh	0
2026	1.46	1.21	1.11	6	0.1 GWh	4
2027	1.83	1.86	1.17	65	6.4 GWh	19
2028	2.23	2.72	1.23	165	31.0 GWh	35
2029	2.66	3.38	1.29	251	68.4 GWh	57
2030	3.08	3.94	1.36	372	117.6 GWh	85

Complete hours of the 2024 record, central NLmin 520 MW and SNSP 75%, before any of the planned battery storage. The SNSP limit first binds in 2029.

> **Why 2028 is the central year**
> Risk is essentially zero until 2026 and becomes material in 2028 (about 31 GWh in the complete hours), the first year the planned solar build-out outpaces demand growth at midday. 2026, 2027 and 2030 are run as the scenario-year sensitivity.

4.9 Charging sessions (SO3)

What

Built `ev_sessions.py`, which creates the set of public charging sessions (arrival, departure, energy, charger power, station) used by every strategy.

How it was done

- **Planned method:** a Gaussian Mixture Model (GMM) fitted to ACN-Data, a public dataset of real charging sessions from Caltech and JPL. It learns arrival hour, plug-in duration and energy *together*; the number of mixture components is chosen by the Bayesian Information Criterion (BIC).
- **Current fallback:** ACN-Data needs a free account that only you can create, so the runs use uniform random sessions, clearly labelled `synthetic_parametric_uncalibrated`.
- **Feasibility cap:** requested energy ≤ charger power × plug-in time. Without it a 100%-completion constraint is impossible. Capped sessions are flagged, not hidden.
- **Stations:** sessions are assigned to stations that share a 50 kW connection.
- **Declared departure:** 80% of drivers state a departure time; the rest cannot be shifted and charge on arrival (the SO6 dimension).
- **Plausibility:** implied driving distance per session at 0.16 kWh/km.

Why this way

Arrival time, dwell time and energy are linked (people who arrive in the morning stay longer). A GMM captures that joint shape, unlike separate distributions.

Alternatives that existed, and why they were not chosen

Alternative	Why not chosen
Separate distribution for each variable	Loses the links between the variables
Copulas	More complex; harder to justify at undergraduate level
Travel-survey models (as Dixon used the UK National Travel Survey)	Sri Lanka has no public travel survey

Earlier research or standard practice followed

- ACN-Data (Lee, Li & Low 2019) and the ACN-Sim simulator (Lee et al. 2021), which fits GMMs to sessions with scikit-learn.
- Monte Carlo sampling of EV load in Sri Lanka (Ranathunga & Samaliarachchi 2017).

How it was validated

- Reproducible (fixed seed); every session on the 15-minute grid; energy never exceeds what can be delivered.
- Plausibility report: mean 17.86 kWh, mean dwell 5.0 h, implied distance median 109.1 km (p95 186.7 km). The uniform fallback gives rather long trips; this is one reason the ACN-Data fit is needed.

Result

Fixing the cap in the original prototype: all 140 sessions now receive their full 2,400.09 kWh. The earlier '161.78 kWh unmet' came from how sessions were generated, not from the charging policy.

4.10 Charging strategies and the LP scheduler (SO4)

`schedule_ev_charging.py` implements three strategies on identical inputs:

Strategy	How it works	Why it is included
Uncontrolled	Charge on arrival at full power; stations share capacity proportionally	Present behaviour; the reference case in Dixon et al. and ACN-Sim
Solar-following	LP that puts energy into the hours with the most available solar	Current 'charge in the daytime' advice; your thesis argues it is mistargeted
Curtailment-aware	LP that minimises unabsorbed curtailment + system peak + driver delay	Your proposed method

The LP formulation

Sets: sessions s with arrival slot a_s, departure slot d_s, energy E_s, charger rating $\bar{P}_s$; 15-minute slots t ($\Delta t = 0.25$ h); hours h; stations k with capacity Cap_k. Decision variables: charging power $p_{s,t}$ (kW), curtailment left c_h (MW), system peak P (MW).

(1) $0 \le p_{s,t} \le \bar{P}_s$ while plugged in ($a_s \le t < d_s$), otherwise 0

(2) $\Sigma_t\, p_{s,t} \cdot \Delta t = E_s$ — full energy is a **hard** constraint

(3) $\Sigma_{s \text{ at } k}\, p_{s,t} + F_{k,t} \le Cap_k$ — F is the fixed load of sessions that cannot shift

(4) $L_h = \Sigma_{t \text{ in } h}\, \Sigma_s\, p_{s,t} / 4000 + L^{fix}_h$ — hourly EV load in MW

(5) $c_h \ge NLmin - NL_h - L_h$; $c_h \ge VRE_h - SNSP \cdot (D_h + L_h)$; $c_h \ge 0$

(6) $P \ge NL_h + L_h$ for every hour

Objective: minimise $\Sigma_h\, c_h + w_{peak} \cdot P + (w_{delay} + \varepsilon) \cdot \Sigma_{s,t}\, p_{s,t} \cdot \Delta t \cdot (t - a_s) \cdot \Delta t / 1000$

> **A technical point you must be able to explain: the epigraph form**
>
> Curtailment left after EV charging is max(0, NLmin − NL − L), which is not linear. It is replaced by a new variable c_h that must be ≥ each expression in the max. Because the LP *minimises* c_h, at the optimum c_h equals the max exactly. This is a standard LP reformulation called the epigraph form. The same trick gives the system peak P.

Why this way

- LP gives a proven optimum and is reproducible.
- Hard constraints guarantee no driver is short-changed for grid benefit (proposal FR10).
- An infeasible day is **reported, not relaxed**: hiding infeasibility would overstate the method.
- A small tie-break term (ε) prefers earlier charging when the grid does not care, so drivers are not delayed for nothing.

Alternatives that existed, and why they were not chosen

Alternative	Why not chosen
Mixed-integer LP (on/off charging)	Only needed for discrete power levels; much slower
Greedy heuristics	No guarantee of the best answer
Reinforcement learning	Needs training data, cannot prove optimality, hard to explain to CEB
Model predictive control (rolling horizon)	A natural later extension once forecasts are added; same LP inside

Earlier research or standard practice followed

- Time-coupled LP scheduling of EV charging against a grid signal (Dixon et al. 2020).
- Offline-optimal LP/convex scheduling with session constraints in ACN-Sim (Lee et al. 2021).

How it was validated

- Solver status must be 'optimal' (HiGHS via SciPy).

- An independent audit re-checks every schedule without trusting the solver: negative power, power above rating, charging while unplugged, over-delivery, station overload, energy not delivered.
- Toy-day test of the core claim: on a day where the sunniest hour (12:00) is not the risk hour (14:00), solar-following absorbs 0 MWh while curtailment-aware absorbs the full 1 MWh.
- Speed: 10,000 sessions per day solve in under 5 seconds (proposal NFR1 requires under 60).

4.11 The forecasting layer: the models that were trained (WBS 5.8)

What

The scheduler needs tomorrow's grid conditions. In reality they are not known in advance, so `forecast_grid.py` trains day-ahead forecasting models and tests how much of the benefit survives when the scheduler plans on a forecast.

How it was done

- **Issue time:** 00:00 for every hour of the coming day; only information known then is used.
- **Targets:** hourly system demand, available solar, wind and must-take generation.
- **Features:** hour, weekday, weekend, Sri Lankan public and Poya holiday flags (known in advance), day after a holiday, season (day-of-year sine/cosine), the value 24 h and 168 h earlier, and yesterday's daily mean.
- **Models:** persistence (yesterday, same hour), seasonal naive (last week), ridge regression, and gradient-boosted trees (HistGradientBoostingRegressor).
- **Selection rule:** a learned model is used only if it beats persistence (proposal Table 3.2).

Why this way

Holidays and weekends are when demand is lowest and curtailment risk is highest, so they are the most important features. Ridge is a transparent linear benchmark; gradient boosting captures non-linear calendar effects.

Alternatives that existed, and why they were not chosen

Alternative	Why not chosen
Deep learning (LSTM, transformers)	Needs far more data than one year; harder to explain; not needed to show forecast value
Statistical time-series models (SARIMA)	Possible; handle holidays poorly without extra regressors
Numerical weather prediction inputs	The best way to improve the solar forecast; a planned extension

Earlier research or standard practice followed

Standard short-term load-forecasting practice: calendar and lag regressors, naive benchmarks, chronological hold-out testing (e.g. Hong & Fan 2016).

How it was validated

- **Chronological hold-out:** train January-September 2024, test October-December 2024 (the model never sees the test months).
- **Decision quality:** precision and recall of risk hours (C(t) > 0) on the test months.
- **Leave-one-month-out** predictions for the scheduling experiment (each month is predicted by a model trained on the other eleven).

4.12 Experiment design (SO5 and SO6)

- **Scenario:** LTGEP base-case year 2028 (solar × 2.23, wind × 2.72, demand × 1.23), NLmin 520 MW, SNSP 75%.
- **Weekly sample:** 39 days, one per week across 2024, rotating the weekday so weekends count fairly. Used for the year-representative evening peak, delay and completion. A day is used only if it, the previous day (for the forecast) and the next morning (for overnight departures) are complete.

- **Risk days:** all 10 usable days with curtailment risk in the central year. Curtailment risk is a rare event, so the weekly sample alone contains only a few of these days; absorption is compared on all of them (a separate stratum, standard for rare outcomes).
- **Strategies:** uncontrolled, solar-following, curtailment-aware with perfect foresight, curtailment-aware planned on the persistence forecast, and curtailment-aware planned on the ML forecast. All are scored on the actual grid.
- **Metrics** (proposal Table 3.3), always reported together: risk energy absorbed and left (MWh), evening peak change (MW), completion rate, mean and 95th-percentile delay, shiftable share, constraint violations, solver status and runtime.
- **Trade-off curve:** peak weight 0-16 × delay weight 0-0.1, on the risk days (where absorption can differ at all).
- **Sensitivity** (on the 42-day weekly sample): NLmin 240/380/520/650/780 MW, SNSP off/0.5/0.65/0.75, scenario year 2026/2027/2028/2030, declared-departure share 50/80/100%. Test: is the strategy ranking preserved? Strategies within 0.5 MWh (absorption) or 0.05 MW (peak) are treated as tied, so solver-level differences are not counted as a change.
- **Computation:** days are solved in parallel worker processes; a test confirms the result is identical to solving them one by one.

5. Results (provisional fleet and session inputs)

> **Read these results correctly**
>
> The grid side (NLmin, SNSP, scenario year) is now cited. The EV side is not: sessions are uncalibrated and the fleet size is a placeholder. The results show how the strategies compare and that every schedule is feasible; absolute MWh and MW will change when ACN-Data and the Component 1 fleet scenario replace the placeholders.

5.1 Absorption on the curtailment-risk days (SO5)

All 10 usable days of the 2024 record that carry curtailment risk in scenario year 2028: 100,000 sessions and 5,265 MWh of risk energy.

Strategy	Absorbed (MWh)	Gain vs uncontrolled	Evening peak increase (MW)	Mean / p95 delay (min)	Completion	Violations
Uncontrolled	445.3	0.0	12.91	0.0 / 0	99.85%	0
Solar-following	512.0	66.8	9.34	44.1 / 180	100.00%	0
Curtailment-aware (perfect foresight)	520.5	75.2	5.16	14.8 / 60	100.00%	0
Curtailment-aware, persistence forecast	453.3	8.0	5.65	9.3 / 60	100.00%	0
Curtailment-aware, ML forecast	477.3	32.0	5.36	12.7 / 60	100.00%	0

Every LP solved to optimality and the independent audit found zero violations. Uncontrolled completion is below 100% because a few sessions arriving together exceed the station's 50 kW connection when everyone charges at once; the scheduled strategies spread them.

Day	Risk (MWh)	Uncontrolled	Solar-following	Curt.-aware	Persistence	ML forecast
2024-02-04	3.1	3.09	3.09	3.09	3.09	3.09
2024-02-11	13.6	12.48	13.56	13.56	12.48	12.48
2024-04-14	721.6	63.81	75.21	75.21	63.81	63.81
2024-06-09	317.1	50.98	59.67	68.11	50.98	60.31

Day	Risk (MWh)	Uncontrolled	Solar-following	Curt.-aware	Persistence	ML forecast
2024-09-17	212.3	37.73	53.72	53.72	45.74	40.39
2024-10-27	843.3	73.38	80.05	80.05	73.38	80.04
2024-11-03	1,395.0	60.48	67.20	67.20	60.48	67.09
2024-11-15	418.5	48.30	55.04	55.04	48.30	55.04
2024-12-01	1,317.0	74.70	81.37	81.37	74.70	74.70
2024-12-29	23.1	20.34	23.12	23.12	20.34	20.34

Absorbed risk energy (MWh) on each risk day (results/experiments/so5_risk_days_by_day.csv).

What the risk days show

- **Curtailment-aware absorbs the most.** It recovers 75.2 MWh more than uncontrolled charging, against 66.8 MWh for solar-following (13% more). The whole difference comes from **one of the ten days**, Sunday 9 June 2024. In the south-west monsoon, wind is about 550 MW all day and demand falls after midday, so the risk peaks at 13:00-14:00 (about 120 MW) while the sunniest hour is 12:00 and 11:00 has no risk at all. Solar-following charges at the sun peak and misses the risk; curtailment-aware follows the risk. On the other nine days the risk sits on the sunny hours and the two absorb the same. This is the proposal's 'solar-following is mistargeted' claim, observed on real grid data, but it is a wind-season effect, not an everyday one.
- **It also cuts the added evening peak by 60%**, against 28% for solar-following, with a mean delay of 15 minutes instead of 44.
- **EV charging absorbs only a small part of the risk.** 10% of the risk energy at 10,000 sessions a day. The risk window is hundreds of MW; the EV load is about 15 MW. Fleet size, not the algorithm, limits the total.
- **Forecasts keep most of the peak benefit but lose much of the absorption.** The ML forecast keeps 43% of the absorption gain and 98% of the peak reduction; persistence keeps 11% and 94%. Nine of the ten risk days follow a day with no risk (mostly a Sunday after a Saturday), so 'same as yesterday' almost never predicts them; the ML model knows the weekday and holidays and does better.

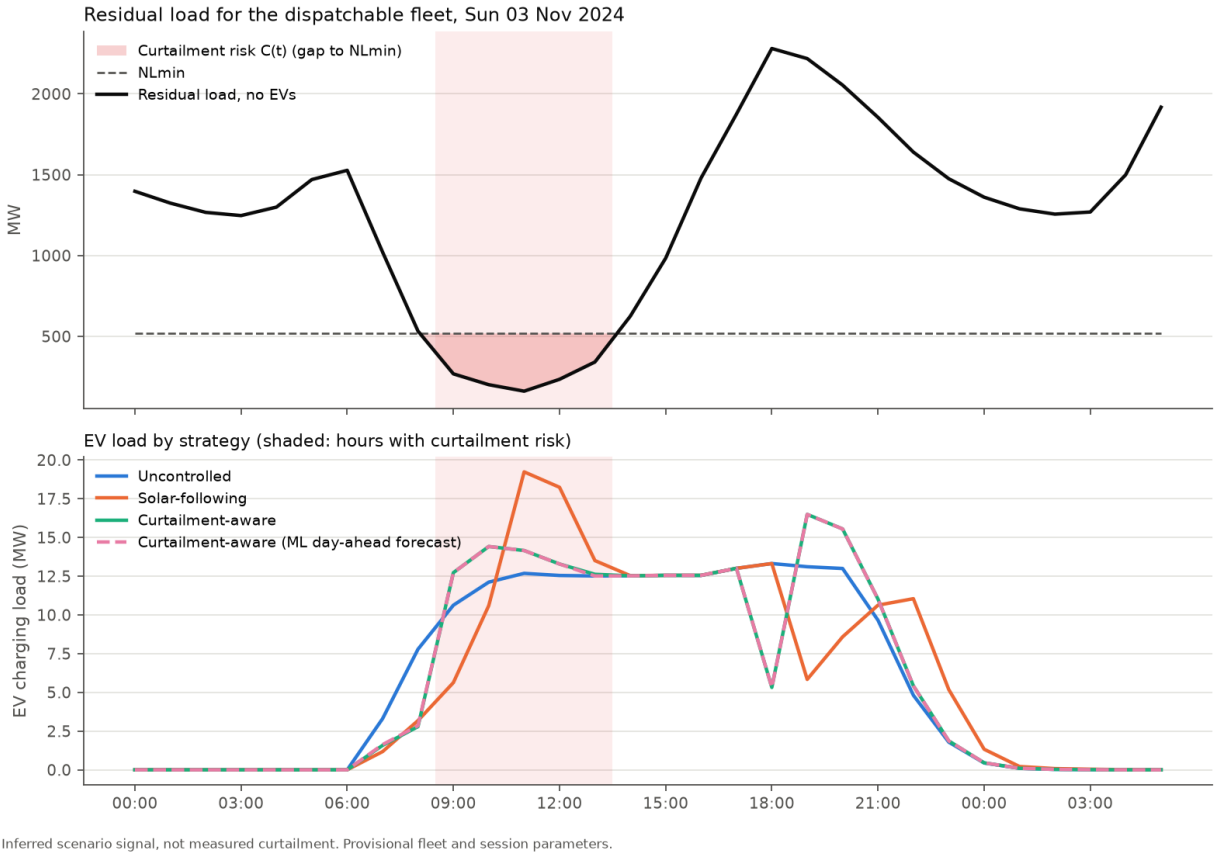


Figure 2. The risk day with the most risk energy. Top: residual load and NLmin; the shaded band is curtailment risk. Bottom: the EV charging load of each strategy. Solar-following concentrates charging at 11:00-12:00; curtailment-aware fills the whole risk window and avoids the 18:00 peak hour.

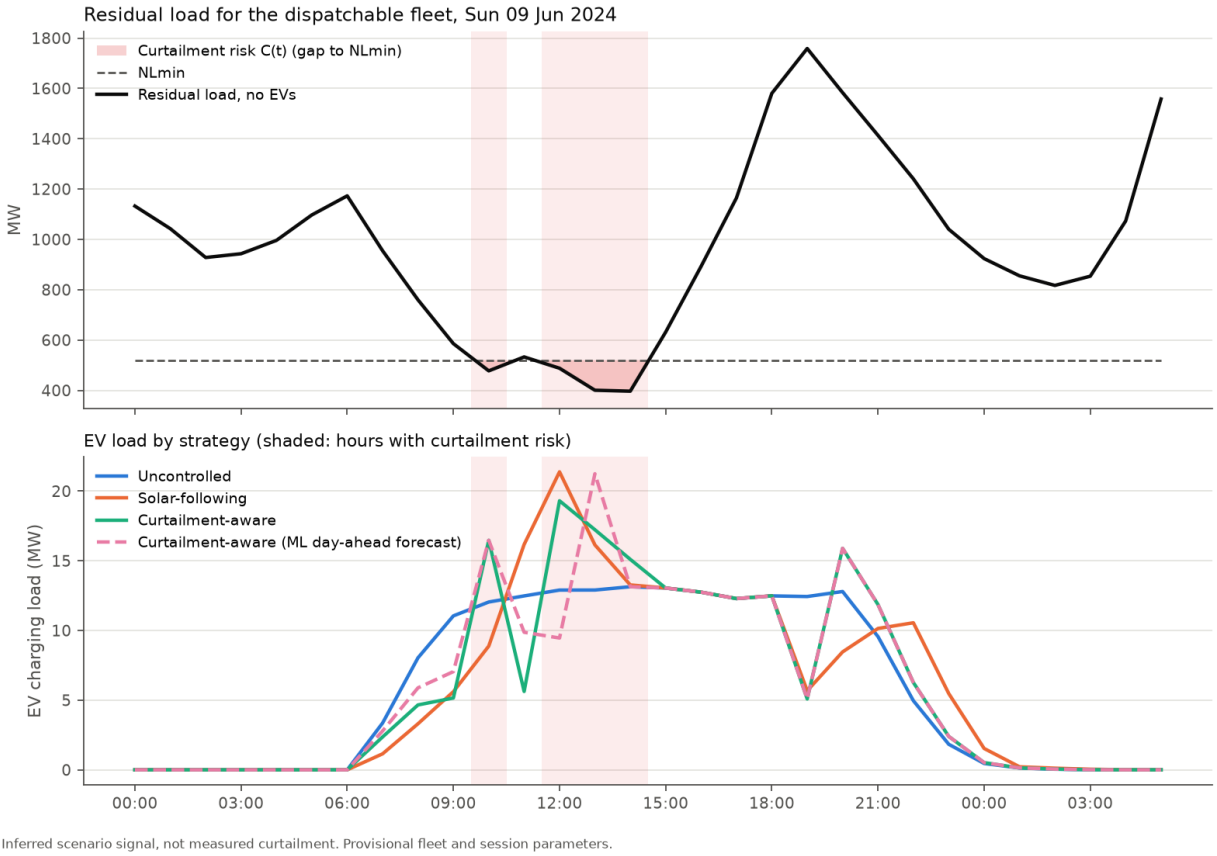


Figure 2b. Sunday 9 June 2024, the risk day where curtailment-aware absorbs the most more than solar-following. Monsoon wind and falling demand put the largest risk at 13:00-14:00, after the 12:00 solar peak; 11:00 has no risk. Solar-following charges at the solar peak; curtailment-aware follows the risk.

5.2 The year-representative sample (SO3, SO5)

One day per week (39 usable days, 390,000 sessions). This shows what happens on ordinary days as well as risk days: the uncontrolled evening peak increase (SO3) and the service cost.

Strategy	Absorbed (MWh)	Gain vs uncontrolled	Evening peak increase (MW)	Mean / p95 delay (min)	Completion	Violations
Uncontrolled	234.1	0.0	12.79	0.0 / 0	99.83%	0
Solar-following	270.2	36.1	8.79	44.5 / 180	100.00%	0
Curtailment-aware (perfect foresight)	270.2	36.1	5.13	8.3 / 60	100.00%	0
Curtailment-aware, persistence forecast	242.1	8.0	6.90	8.7 / 60	100.00%	0
Curtailment-aware, ML forecast	250.2	16.1	5.79	8.6 / 60	100.00%	0

3 sampled day(s) were dropped for all strategies because an LP was infeasible (2024-03-02, 2024-07-13, 2024-10-11); they are reported, not relaxed.

- **Uncontrolled charging adds 12.8 MW** to the evening peak on an average day at 10,000 sessions/day (SO3).
- Absorption is equal for solar-following and curtailment-aware on this sample (270.2 vs 270.2 MWh): the few risk days in it have midday risk, where both strategies charge.
- Curtailment-aware still removes 60% of the added evening peak (solar-following 31%), with a mean delay of 8 minutes (solar-following 44).

How to state the main finding

On the days with curtailment risk, curtailment-aware scheduling absorbs 13% more at-risk renewable energy than solar-following (all of it on a monsoon day when wind moved the risk away from the solar peak), removes 60% of the evening peak that EV charging adds (solar-following 28%), and delays drivers by 15 minutes on average instead of 44.

Do not overstate the absorption difference: it is real but modest, and on days when the risk is at the sunniest hours the two strategies absorb the same. The larger and more robust difference is peak and driver delay. This is exactly why the proposal requires grid and service metrics to be reported together.

5.3 Forecast accuracy (models trained)

Target	Persistence MAE	Seasonal naive	Ridge	Gradient boosting	Selected
System demand	132.3 MW	122.7 MW	91.2 MW	75.2 MW	gbm
Available solar	23.6 MW	38.4 MW	24.7 MW	29.4 MW	persistence
Wind	34.6 MW	62.0 MW	31.6 MW	34.9 MW	ridge
Must-take	25.8 MW	52.4 MW	22.7 MW	33.8 MW	ridge

Hold-out October-December 2024. Gradient boosting improves demand forecasts by 43% over persistence (MAE 75 vs 132 MW). No learned model beats persistence for solar without weather-forecast inputs, so persistence is kept for solar - exactly as the selection rule requires.

Risk-hour detection, Oct-Dec, scenario 2028	Precision	Recall	Forecast vs actual risk energy
Persistence	0.17	0.148	9,648 vs 12,627 MWh
Selected models	0.705	0.508	7,988 vs 12,627 MWh

Precision: of the hours the forecast flags as risky, the share that really were. Recall: of the really risky hours, the share the forecast found. The learned models find risk hours far more reliably than persistence but under-estimate how large the risk is, because they smooth extremes.

5.4 Trade-off between absorption, peak and delay (SO5)

Peak weight	Delay weight	Absorbed (MWh)	Peak reduction (MW)	Mean delay (min)
0.0	0.0	520.5	0.00	7.3
0.25	0.0	520.5	7.75	14.9
1.0	0.0	520.5	7.75	14.9
4.0	0.0	520.5	7.75	14.9
16.0	0.0	520.5	7.75	14.9
0.0	0.01	520.5	0.00	7.2
0.25	0.01	520.5	7.75	14.8
1.0	0.01	520.5	7.75	14.8
4.0	0.01	520.5	7.75	14.8
16.0	0.01	520.5	7.75	14.8
0.0	0.1	520.5	0.00	7.2
0.25	0.1	520.5	5.56	12.6
1.0	0.1	520.5	7.75	14.8
4.0	0.1	520.5	7.75	14.8
16.0	0.1	520.5	7.75	14.8

On the risk days, absorption is exactly 520.5 MWh at every one of the 15 weight settings. The evening peak reduction ranges from 0.00 to 7.75 MW and the mean delay from 7 to 15 minutes.

- **There is no conflict between absorbing curtailment and cutting the evening peak.** Both ask for the same thing: move charging out of the evening and into the midday risk window. The EV load (about 15 MW) is far smaller than the risk window, so the scheduler can do both at once.
- **The real trade-off is evening peak against driver delay.** With peak weight 0 the scheduler has no reason to avoid the evening, so it gives no peak reduction at about 7 minutes of delay. Any peak weight from 0.25 upwards gives the full reduction at about 15 minutes. A high delay weight (0.1) with a small peak weight (0.25) stops in between.
- So the central weights (curtailment 1, peak 1, delay 0.01) lie on the flat part of the curve: the result does not depend on fine-tuning them. The proposal expected a trade-off curve; the honest finding is that, at this fleet size, the curve is a step.

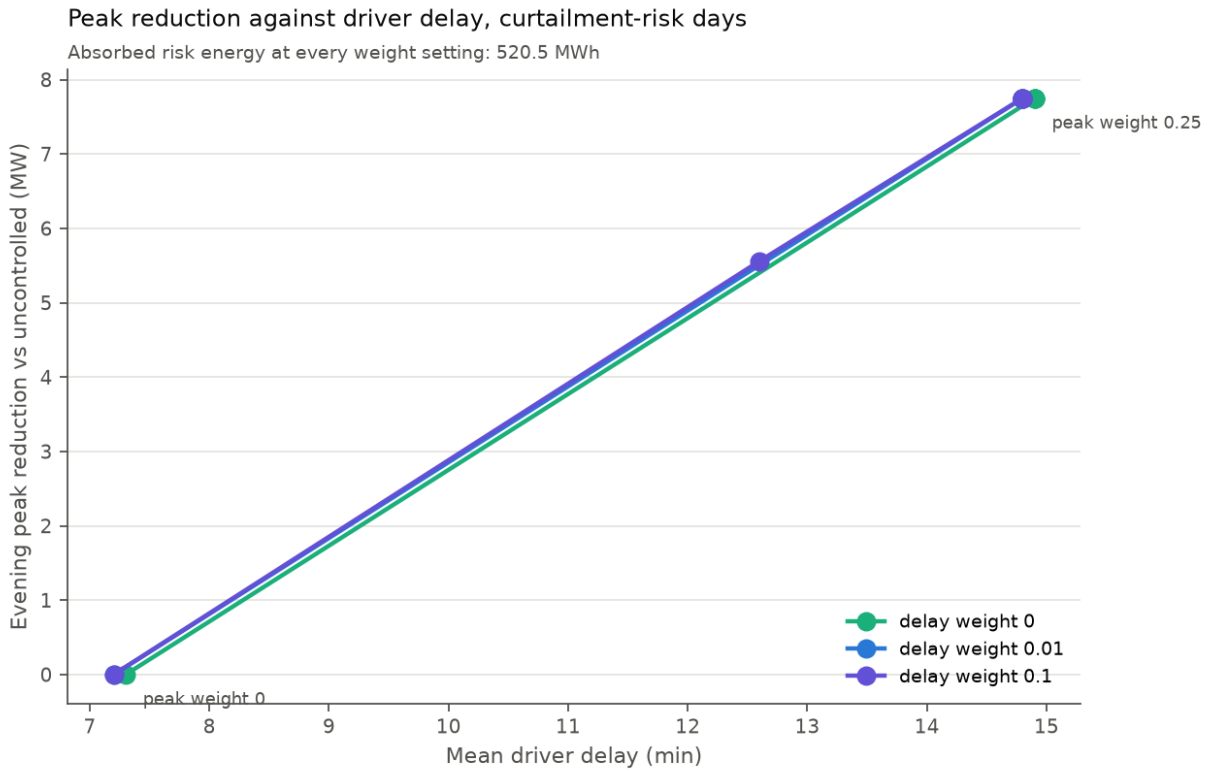


Figure 3. Evening peak reduction against mean driver delay when the peak weight is swept from 0 to 16 at three delay weights, on the curtailment-risk days. Absorption does not change between settings, so it is stated above the plot rather than drawn as an axis.

5.5 Sensitivity: is the ranking preserved? (SO6)

Dimension	Value	MWh U	SF	CA	Peak MW U	SF	CA	Days	Exact ranking kept	CA never outranked
NLmin (MW)	240	25.4	38.1	38.1	12.79	8.79	5.13	39	yes	**yes**
NLmin (MW)	380	101.3	125.3	125.3	12.79	8.79	5.13	39	yes	**yes**
NLmin (MW)	520	234.1	270.2	270.2	12.79	8.79	5.13	39	yes	**yes**
NLmin (MW)	650	284.0	325.2	325.2	12.79	8.79	5.13	39	yes	**yes**
NLmin (MW)	780	379.0	409.8	428.4	12.79	8.79	5.13	39	no	**yes**
SNSP limit	off	234.1	270.2	270.2	12.79	8.79	5.13	39	yes	**yes**
SNSP limit	0.5	465.4	546.9	548.8	12.79	8.79	5.13	39	no	**yes**
SNSP limit	0.65	234.1	269.7	270.2	12.79	8.79	5.13	39	yes	**yes**

Dimension	Value	MWh U	SF	CA	Peak MW U	SF	CA	Days	Exact ranking kept	CA never outranked
SNSP limit	0.75	234.1	270.2	270.2	12.79	8.79	5.13	39	yes	**yes**
Scenario year	2026	0.0	0.0	0.0	12.79	8.75	5.13	39	no	**yes**
Scenario year	2027	70.2	80.7	86.6	12.79	8.66	5.13	39	no	**yes**
Scenario year	2028	234.1	270.2	270.2	12.79	8.79	5.13	39	yes	**yes**
Scenario year	2030	469.0	543.7	550.0	12.79	8.79	5.13	39	no	**yes**
Declared departure share	0.5	159.4	178.2	178.2	12.85	10.36	8.03	22	yes	**yes**
Declared departure share	0.8	234.1	270.2	270.2	12.79	8.79	5.13	39	yes	**yes**
Declared departure share	1.0	234.1	279.4	279.4	12.79	7.75	3.18	42	yes	**yes**

U = uncontrolled, SF = solar-following, CA = curtailment-aware; absorbed in MWh, peak increase in MW; 42-day weekly sample. Reference ranking (central case): by absorption curtailment_aware = solar_following > uncontrolled; by evening peak curtailment_aware < solar_following < uncontrolled.

- **The exact ranking is kept in 11 of 16 settings, and curtailment-aware is never outranked in 16 of 16.** Every exact-ranking 'change' is curtailment-aware moving from *tied with* solar-following to *ahead of* it (NLmin 780 MW, SNSP 50%, scenario years 2027 and 2030), or all three tying at zero because there is no risk (2026). No setting reverses the order. Reporting only 'ranking preserved' would hide this, so both measures are reported.
- **Absorption grows with tighter grid limits and later years** (for example 38 MWh at NLmin 240 MW to 428 MWh at 780 MW). The curtailment-aware advantage over solar-following appears exactly where risk is larger and spreads beyond the solar peak: +18.6 MWh at NLmin 780 MW, +5.9 MWh in 2027, +6.3 MWh in 2030.
- **The evening-peak ranking (CA < SF < U) holds in every setting.** The peak results barely depend on the grid parameters, because they are driven by when drivers arrive.
- **Declared departure share is the most sensitive input.** At 100% declaration every day is feasible and curtailment-aware cuts the added evening peak to 3.2 MW. At 80% three of the 42 days are infeasible, and at 50% twenty are: the undeclared drivers charge on arrival, fill the shared 50 kW station connection, and leave no room to deliver the declared drivers' full energy. These days are reported, never relaxed. In practice this argues for making departure declaration the default in the charging app, or for a station-level priority rule.

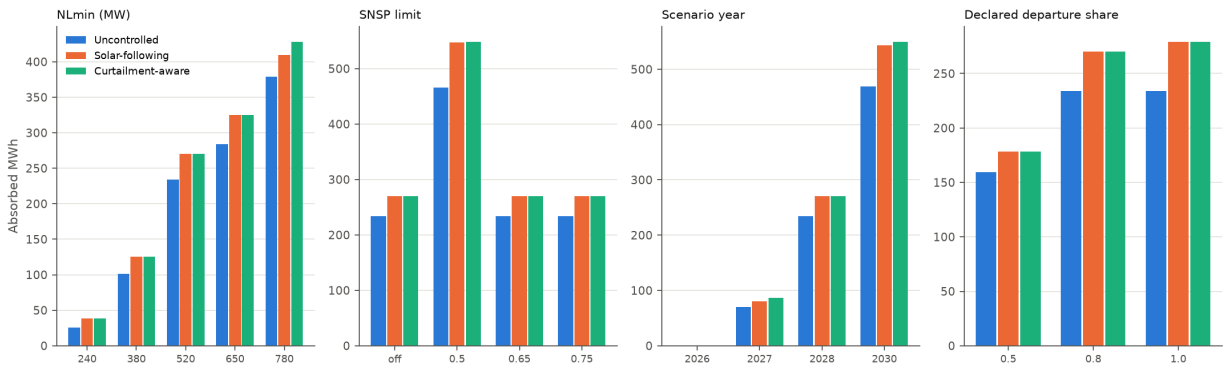


Figure 4. Absorbed risk energy for each strategy across the sensitivity dimensions (weekly sample).

5.6 Scalability and speed (NFR1, NFR2)

Sessions per day	Stations	LP variables (session × slot)	Solve time	Status	Violations
1,000	125	19,923	0.2 s	optimal	0

Sessions per day	Stations	LP variables (session × slot)	Solve time	Status	Violations
3,000	375	59,759	0.8 s	optimal	0
10,000	1250	199,140	4.7 s	optimal	0
30,000	3750	600,470	31.8 s	optimal	0

`scripts/benchmark_scaling.py`, one process, on the hardest risk day (2024-11-03). The central volume (10,000 sessions) solves in 4.7 s, far inside the 60 s of NFR1. Over a 30-fold range of session counts (NFR2 asks for at least 10-fold) every solve is optimal with zero violations, and runtime grows roughly as sessions$^{1.5}$: a sparse LP grows a little faster than linearly. About 45,000 sessions per day would still fit within 60 s; beyond that, days or regions can be solved in parallel, as the experiments already do.

6. Which models exist, and which were trained

Be precise about this in your viva. The core of the method is **optimisation** and a **physical model**, which are solved or computed, not trained. Machine-learning models are used for forecasting and, once data is available, for session generation.

Component	Type	Trained?	On what data	Validated by
Day-ahead demand forecast	Gradient-boosted trees (ML)	Yes	2024 hourly system demand + calendar/holiday features	Chronological hold-out; skill vs persistence
Wind and must-take forecasts	Ridge regression (ML)	Yes	Same	Same
Solar forecast	Persistence benchmark	No (no learned model beat it)	-	Same
Session model	Gaussian Mixture Model (ML)	Code ready; not yet trained	ACN-Data, once downloaded	BIC model selection; plausibility check
Available solar	Physical scaling of irradiance to official energy	No (calibrated)	NASA POWER + CEB monthly	Correlation with metered solar; energy match
C(t)	Rule-based physical model	No	-	Signal plausibility; cross-check
LP scheduler	Mathematical optimisation	No (solved)	-	Solver status; independent audit; tests

> **If your panel expects more machine learning**
>
> The natural extension is a stronger forecasting layer: add numerical weather-prediction inputs for solar, and probabilistic forecasts (quantiles) so the scheduler can plan for uncertainty. Both fit inside WBS 5.8 without changing the research question.

7. How every result is validated

Level	Check	Evidence
1. Data	Quality flags; no imputation; coverage counts	99.73% coverage; 1,263 flagged intervals visible
2. Arithmetic	Hourly totals equal daily totals	Exact on 255 complete days
3. External	Reconcile with CEB published totals	Demand within 0.1%; coal within 0.2%
4. Physical	Solar zero at night; shape matches metered solar; identity at today's solar	Correlation 0.963; night 0 MW
5. Signal	Risk at midday, larger on weekends; NLmin cross-check with observed operation	89% of risk 10:00-15:00; weekend ≈ 9 × weekday; 0 observed hours below NLmin
6. Model	Optimal solver status; independent constraint audit; automated tests	Zero violations on every run; 159 tests pass
7. Result	Identical inputs for every strategy; ranking preserved across sweeps; infeasible days reported	Curtailment-aware never outranked in 100% of 16 settings; exact ranking kept in 69% (the rest are ties becoming wins)
8. Forecast	Chronological hold-out; skill vs persistence; risk-hour precision/recall	Section 5.2
9. Reproducibility	Fixed random seeds; all assumptions in one config file; git version recorded in each run	config/scenarios.json; run_summary.json
10. Expert review	CEB System Control reviews NLmin derivation	Still to do (proposal §3.5)

8. Every change made on 19 October 2026

8.1 Corrections to existing work

File	Change	Reason
scripts/ev_charging_baseline.py	Energy capped at charger power × plug-in time; capped sessions flagged	19 of 140 sessions were impossible to fully charge
docs/nlmin_derivation.md	Updated definitions, numbers, SNSP option and hourly cross-check	Still said 2025 was 'unaudited' and quoted superseded numbers
docs/data_inventory.csv	Added PUCSL, CEB Data Book, NASA 2024 (3 sites), ACN-Data status	Primary dataset was missing from the inventory
README.md, config/project.json	Test count, pipeline commands, project status	Out of date

8.2 New components

File	Purpose
config/scenarios.json	Every assumption in one place; unsourced values tagged 'provisional'
scripts/build_grid_scenario.py	Boundary correction, available solar, NL(t), C(t) (SO1, SO2)
scripts/ev_sessions.py	Session sets: parametric fallback and ACN-Data GMM (SO3)
scripts/schedule_ev_charging.py	Strategies, LP, audit, metrics (SO4)
scripts/forecast_grid.py	Day-ahead forecasting models (WBS 5.8)
scripts/run_experiments.py	SO5 comparison, trade-off, SO6 sensitivity, forecast planning
docs/methods_and_literature.md	Method-to-precedent map, deviations, novelty
scripts/derive_nlmin.py	NLmin from LTGEP coal minimum + observed commitment, with reserve, 25%-unit and cross-checks (SO2, Appendix C)
build_grid_scenario.ltgep_scenario()	LTGEP scenario years 2025-2030: solar, wind, must-take, demand

File	Purpose
app/streamlit_app.py, app/app_pages/, scripts/dashboard_data.py	User-friendly prototype (WBS 5.11, FR13): plain-language pages, ready-made days, a written verdict, a driver's view; the scheduler runs live. Earlier: interactive dashboard (WBS 5.11, FR13): pick a day and settings, run the scheduler live; saved results; scenario years; assumptions
tests/test_*.py (8 new files)	56 new tests; 159 in total

8.2a Second round of changes (same day, after review of the first results)

Change	Why
NLmin 700 MW placeholder → 520 MW derived from LTGEP Table 2.7 and p. 7-13 plus the 2024 stressed hours; Appendix C filled in	SO2 requires a documented, cited derivation; 700 MW was crossed by observed operation in 32 hours
SNSP off → 75% (LTGEP)	CEB plans with this limit; it is the cited inertia constraint
Solar × 1.5 → LTGEP scenario year 2028 (solar, wind, must-take and demand scaled)	Uncited multiplier; ignoring demand growth overstated 2028 risk four times
Second evaluation set: every risk day	With cited values risk is rare; the weekly sample held only a few risk days
Sweeps on all 42 days instead of every 4th	The 11-day subset had no risk; strategies tied trivially
Tie-aware ranking (0.5 MWh / 0.05 MW)	A 0.02 MWh solver difference counted as a ranking change
Two-panel day figure; hourly EV loads saved; second figure for the day with the largest curtailment-aware advantage	EV lines overlapped and hid each other
Trade-off figure redrawn as peak reduction vs delay	Absorption did not change with the weights, so an absorption axis showed a single vertical line
'Never outranked' robustness measure next to 'ranking preserved'	Ties turning into wins were being counted as ranking changes
Parallel day solving (--workers)	Full sweeps in minutes instead of hours; same results (tested)

8.3 Deviations from the approved proposal (need supervisor approval)

#	Proposal said	Now	Why
D1	CEB hourly demand data	PUCSL 15-minute data reconciled to CEB monthly totals	CEB publishes no hourly archive
D2	Net demand from metered generation	Available solar modelled from NASA POWER	Metered output hides curtailment
D3	NLmin = sum of minimum generation + reserve + inertia	NLmin = coal units × 150 MW + hydro kept online = 520 MW; reserve as a check; inertia as the LTGEP 75% SNSP limit	Avoids double-counting; inertia is not MW; CEB states inertia as SNSP
D4	Hydro dispatchable	Major hydro dispatchable; mini hydro and biomass must-take	Mini hydro is run-of-river
D5	Session energy unconstrained	Capped at deliverable energy	Otherwise 100% completion is impossible
D6	PuLP/Pyomo + HiGHS	SciPy linprog + HiGHS	Same solver; much faster to build
D7	Evidence year 2024	2024 now; 2025 recommended	Real curtailment began in 2025
D8	Forecasting layer in Feb-Mar 2027	Day-ahead ML forecast now	Needed to measure forecast value
D9	Solar growth scenarios	LTGEP base-case years 2025-2030; central 2028	Cited; demand grows with solar

#	Proposal said	Now	Why
D10	One evaluation set	Weekly sample + every risk day	Risk is a rare event

None of these changes the research topic, the main objective or SO1-SO6. They change *how* the objectives are met, and each is justified by data or by published practice. Methods are expected to evolve during research; what matters is that every change is logged, justified, approved and reported in a 'Deviations from the proposal' section of the methodology chapter.

8.4 Problems found in the proposal document itself

- In-text citation numbers do not match the reference list ([5], [6], [7] point to the wrong entries; ACN-Data cited as [4] but listed as [7]; Jayasinghe cited as [2] but listed as [3]).
- Ranathunga & Samaliarachchi (2017) is cited but missing from the reference list; reference [10] does not exist.
- The list of appendices (A-G) does not match the appendices included; the NLmin worksheet promised as Appendix C is not included.
- **The AI-use disclosure describes a different project** (Set Covering, OSMnx, Greedy Heuristic). This is an academic-integrity risk and must be corrected by you.
- Template instructions left in the text ('[Verify volume, page range and DOI]').
- Uncited claims: '40 to 50 percent minimum load', and that island curtailment arises 'principally from minimum generation and inertia'.
- The research-gap sentence 'no reviewed study derives the signal indirectly' is too strong: Denholm & Margolis (2007) did. Narrow it as in Section 10.

9. Earlier research followed, and what was taken from each

Source	What it did	What this research takes from it
Dixon, Bukhsh, Edmunds & Bell (2020), *Renewable Energy* 161	Scheduled EV charging by a time-coupled linearised optimal power flow against measured GB carbon and wind-curtailment data	Controlled-experiment design; uncontrolled baseline; LP-type scheduling; the gap (their signal is measured)
Denholm & Margolis (2007), *Energy Policy* 35(5)	Simulated hourly PV and found it unusable when inflexible conventional plants cannot turn down	Inferring curtailment from a minimum-generation floor; simulating available PV
EirGrid operational constraints (Ireland)	Runs an island grid with an SNSP limit and a minimum number of synchronous units	The inertia constraint form (SNSP)
Lee, Li & Low (2019), ACN-Data	Open dataset of real EV charging sessions	Session data source
Lee et al. (2021), ACN-Sim	Simulator with GMM session generation and LP/convex schedulers	GMM session model; scheduling benchmarks
Jayasinghe, Wijekoon & Ekanayake (2018)	Reserve needed for EV charging in Sri Lanka	Reserve component of NLmin (Appendix C)
Ranathunga & Samaliarachchi (2017)	Monte Carlo EV load on the Sri Lankan load profile	Local precedent for probabilistic session sampling
Hong & Fan (2016), *Int. J. Forecasting* 32(3)	Review of electric load forecasting	Calendar/lag features, naive benchmarks, hold-out testing
CEB Long Term Generation Expansion Plan 2025-2044	The operator's 20-year plan: plant data, planning rules, capacity additions, demand forecast, an aggregate EV scenario	Coal minimum, reserve, 25% rule, SNSP 75%, scenario years; the planning-level EV result this work tests at session level
CEB Sales and Generation Data Book 2024; NASA POWER; PUCSL reports	Official data	Reconciliation, irradiance shape, dispatch series

Verify every bibliographic detail (volume, pages, author list) against the original before citing. The ACN-Sim journal version and author list in particular should be checked.

10. What is new in this research (novelty)

10.1 Not new - cite these as foundations

- Inferring curtailment from minimum-generation constraints (Denholm & Margolis 2007).
- LP scheduling of EV sessions against a grid signal (Dixon et al. 2020; ACN-Sim).

10.2 New in this component

1. **Scheduling public EV charging against an inferred, constraint-derived curtailment signal** in a system that publishes no curtailment record. Prior curtailment-aware scheduling uses measured curtailment.
2. **An island-system formulation of the signal** - minimum synchronous generation plus an SNSP-style inertia limit, rather than oversupply or congestion - matching the stated cause of Sri Lanka's 2025 curtailment events.
3. **A measured test of the 'solar-following is mistargeted' claim**, reporting absorption, peak and driver delay together on identical sessions.
4. **Forecast value**: how much of the benefit survives with a day-ahead forecast instead of perfect knowledge.
5. **A reconciled Sri Lankan grid dataset**: the first documented reconciliation of PUCSL 15-minute dispatch with the CEB Data Book, showing that PUCSL excludes rooftop and embedded generation, with corrected demand matching CEB net generation within 0.1%.

10.3 Position against CEB's own EV study

The LTGEP 2025-2044 contains an 'Aggressive EV Scenario' (§10.17, p. 10-39; Fig. 10.28, p. 10-40). It adds an **aggregate** hourly EV profile to demand and shifts a flexible share with time-of-use tariffs, and it finds that daytime charging reduces renewable curtailment. Your examiners or CEB may ask how your work differs. It differs in three ways:

- It schedules **individual sessions** under arrival, departure, energy, charger and station limits, not an aggregate profile.
- It targets an **hour-by-hour risk signal** C(t), not a fixed time-of-use window.
- It reports the **driver-side cost** (delay, completion) next to the grid benefit.

Cite the LTGEP result as the planning-level motivation; your component tests whether the benefit holds at session level, and what it costs drivers.

> **Suggested research-gap sentence**
> "Curtailment can be inferred from minimum-generation constraints (Denholm & Margolis, 2007), and EV charging has been scheduled against measured curtailment (Dixon et al., 2020), but no study schedules public EV charging against an inferred curtailment signal in an island system where curtailment is driven by minimum synchronous generation and inertia limits and no curtailment record is published."

11. Limitations and threats to validity

Limitation	Effect	How it is handled or will be
NLmin derived from public values, not reviewed by CEB	Absolute risk energy uncertain	Swept 240-780 MW; 0 observed hours below 520 MW; CEB review planned
2025 operation may have used a higher floor (post-blackout)	Central case may under-state 2025-26 risk	650 and 780 MW sweep points; test with PUCSL 2025 data
Planned battery storage not modelled	Risk is before storage; EV benefit overstated where batteries would absorb it	State as scope; storage is a competitor, not a substitute, for the analysis
2024 weather reused for future years	One weather year	Add 2025 NASA data; note as limitation
Sessions uncalibrated (uniform)	Absorption and peak numbers not realistic	Fit GMM to ACN-Data; transferability sensitivity
ACN-Data is US workplace charging	May not match Sri Lankan public charging	Treat parameters as scenario variables; seek local operator data
Fleet volume is a placeholder	Scales all EV results	Component 1 scenarios
2024 has little real curtailment	Signal cannot be validated against real events	Add 2025 data; test C(t) against known curtailment days
CEB rooftop figure may be exports only	Underestimates rooftop solar	Confirm definition with CEB
Wind curtailment not modelled	Small; metered wind used as available	Note in dissertation
Solar forecast has no weather inputs	Forecast value understated	Add NWP inputs
Distribution network limits not modelled	Local overloads ignored	Station capacity only; state as scope boundary

12. What to do next

#	Task	Who	When
1	Correct the AI-use disclosure in the proposal (your own statement)	You	Before 23 Oct
2	Present deviations D1-D10 to your supervisor as 'the data showed X, so the method became Y'	You	Before 23 Oct
3	State the main result correctly (Section 5)	You	Presentation
4	Decide on adding 2025 data; approve D1-D10; agree how much ML is expected	Supervisor	This week
5	Register for ACN-Data; download sessions JSON to data/raw/charging_sessions/	You	This week
6	Download PUCSL April-December 2025 workbooks; test C(t) against the known 2025 curtailment days	You	This week
7	Send Appendix C (NLmin 520 MW) to CEB System Control (Mr. Lasantha Prasad) for review; ask whether 2 coal units + ~220 MW hydro matches their practice and what floor they used after February 2025	You	This week
8	Ask Component 1 for public-session volume per scenario year (2026-2030)	You	This week
9	Re-run the pipeline with real inputs; update the research log	You (with tools)	After 5-8
10	Expert review of the dashboard and NLmin (WBS 5.12)	You	Apr 2027

13. Progress Presentation 1: evidence for each marking criterion

This section maps the work to the RP (IT4010) Progress Presentation 1 mark sheet. For each criterion it lists what to show, where the evidence is, and what is still missing. Weights are the sheet's: LO1 10%, LO2 25%, LO3 40%, LO4 15%, LO5 10%.

Criterion (weight)	What to show	Evidence in the project	Gap / action before PP1
LO1 Problem definition (30% of 10%)	The problem, tied to what is implemented	2025 curtailment events; midday-dip analysis (16.7% to 61.1% of days); scenario-year table: risk ≈ 0 in 2025, 31 GWh in 2028, 118 GWh in 2030 (Section 4.8b)	None. Lead with the scenario-year table: it shows the problem growing in CEB's own plan.
LO1 Proof of concept (70% of 10%)	The proposed solution working on real data	End-to-end pipeline on real 2024 data; Figure 2b (9 June: curtailment-aware follows the risk, solar-following misses it); 0 violations in every run	Demo the dashboard live (`streamlit run app\streamlit_app.py`): pick 9 June 2024 and show solar-following missing the risk. Have the figures ready in case the demo fails.
LO2 Key pillars (30% of 25%)	The right knowledge areas, applied	Data engineering (reconciliation within 0.1%); operations research (LP); machine learning (forecasting with hold-out); statistics (stratified evaluation, sensitivity); power-system domain (NLmin, SNSP from the LTGEP)	Name these five pillars on one slide and point each to a result.
LO2 Technologies (70% of 25%)	Technologies used and in-depth knowledge	Streamlit + Altair (interactive dashboard); Python, pandas, NumPy; SciPy linprog + HiGHS with sparse matrices; scikit-learn (Ridge, HistGradientBoosting); holidays (LK Poya); matplotlib; pytest; git; parallel processes	Be ready to explain: the epigraph form; why an LP (proven optimum); HiGHS; why chronological hold-out; why persistence is the benchmark (Appendix D).
LO3 Design excellence (20% of 40%)	Sound design	Modular pipeline, each stage a tested script joined by files (Figure 1); all assumptions in one config file with sources; one interface for parametric and GMM sessions; independent audit	Show Figure 1, one line of config/scenarios.json with its source, and the dashboard: a thin UI over a tested data layer that reuses the experiment code.
LO3 Completion (30% of 40%)	About 50% done; no delay against the plan	WBS 5.1-5.10 implemented (Section 3), ahead of the Oct 2026 - Jun 2027 plan; 5.11-5.13 not started, as planned	Say clearly what is provisional (sessions, fleet) and the corrective action for each (risk register below).
LO3 Standards / best practices (20% of 40%)	Evidence of standards	159 automated tests; fixed seeds; git commit recorded in every run; never-impute data rule; official cited sources; chronological hold-out; persistence benchmark; infeasible days reported, not relaxed	Show the test run (159 passed) and run_summary.json.
LO3 User / functional requirements (20% of 40%)	Realistic requirements, well described	Proposal Table 5.1 (FR1-FR14) and 5.2 (NFR1-NFR8); traceability matrix below: 13 of 14 FR done (FR7's per-session export still to add), 1 open; 6 of 8 NFR met, 2 partly met	Show the matrix; FR12 (export scheme) needs the group's agreed schema.
LO3 Risk mitigation (10% of 40%)	Risks with actions being executed	Risk register below	Show which actions are already done (fallbacks, sweeps, benchmark).
LO4 Communication (15%)	Structure, flow, Q&A	Slide outline below; Appendix D viva answers	Rehearse with a timer.

Criterion (weight)	What to show	Evidence in the project	Gap / action before PP1
LO5 Commercialisation (10%)	Business potential with achievable user benefits	Proposal Section 6 (CEB planners, PUCSL, charging operators; free, professional, consultancy and institutional tiers). Measured benefits: 60% of the EV-added evening peak removed, at a mean delay of about 15 minutes, with more at-risk renewable energy absorbed	Turn the MW of evening peak avoided into a value using a published CEB cost (e.g. peaking-plant energy cost). Do not invent a figure.

13.1 Requirements traceability (proposal Tables 5.1 and 5.2)

ID	Requirement (short)	Status	Where / evidence
FR1	Ingest, time-align, validate net demand	Done	preprocess_pucsl_dispatch.py, build_grid_scenario.py; 0.1% vs CEB
FR2	NLmin from documented parameters, adjustable	Done	derive_nlmin.py; config; Appendix C
FR3	Hourly C(t); curtailment and peak windows for any day	Done	scenario_signal(): curtailment_risk_mw, is_peak_window
FR4	Session set from configurable distributions and fleet	Done (parametric); GMM ready	ev_sessions.py; needs ACN-Data
FR5	Uncontrolled baseline and metrics	Done	asap_profile(); so5 tables
FR6	Solar-following comparator	Done	solve_lp(mode='solar_following')
FR7	Optimised schedule per session per interval	Done; export open	15-min power matrix per session; write it out with FR12
FR8	Arrival, departure, energy, charger, station; 100% energy	Done	Hard LP constraints; audit; infeasible days reported
FR9	Grid and service metrics together	Done	so5_comparison.csv, so5_risk_days_comparison.csv
FR10	Trade-off curve from weight sweep	Done	so5_tradeoff.csv; Figure 3
FR11	Sensitivity: NLmin and declared share	Done (+ SNSP, year)	so6_sensitivity.csv; Figure 4
FR12	Export in the group-agreed scheme	**Open**	Needs the group schema
FR13	Dashboard view	Done	app/streamlit_app.py: wasted-power hours and peak marked, options on common axes, driver timeline, strategies on common axes, trade-off, metrics table
FR14	Record assumptions and parameters of every result	Done	run_summary.json: config + git commit
NFR1	Single day at national volume < 60 s, optimal	Met	4.7 s at 10,000 sessions
NFR2	Runtime growth over ≥ 10× session range	Met	1,000-30,000 sessions (Section 5.6)
NFR3	Audit zero violations; optimal status	Met	Every run; solver status recorded
NFR4	Regenerable with fixed seeds	Met	Two full runs gave identical results
NFR5	Planner understands the main result unaided	**Partly**	Dashboard built; needs expert review
NFR6	Assumptions visible and adjustable	Met	config/scenarios.json with sources
NFR7	No personal data	Met	Aggregate and synthetic data only
NFR8	Accept measured curtailment / session data without restructuring	**Partly**	Sessions: yes (GMM interface). Measured curtailment as a signal source: still to add

13.2 Risk register

#	Risk	Impact	Mitigation already in place	Next corrective action
R1	No local session data; ACN-Data needs registration	High	Labelled parametric fallback; GMM code and tests ready	Register and download this week
R2	NLmin not reviewed by CEB	High	Derived from cited LTGEP values; 0 observed hours below it; swept 240-780 MW	Send Appendix C to CEB System Control
R3	Fleet volume unknown (Component 1)	Medium	Results per 10,000 sessions/day; benchmark to 30,000	Agree volumes per scenario year
R4	2024 has no measured curtailment to validate C(t)	Medium	LTGEP scenario years; signal plausibility checks	Download PUCSL 2025; test against known 2025 events
R5	Few drivers declare departure: infeasible days	Medium	Infeasible days reported, never hidden; swept 50-100%	Test a station priority rule
R6	Solver too slow for large fleets	Low	Measured: 30,000 sessions in 32 s; days solved in parallel	None needed now
R7	Deviations D1-D10 not approved	Medium	Each one documented and justified	Present to supervisor before 23 Oct
R8	Group integration (FR12, FR13) late	Medium	Outputs already in CSV/JSON	Agree the schema with the group
R9	AI-assistance disclosure wrong in the proposal	High	Assistance documented (Section 0.1)	Correct the disclosure; be able to explain all code

13.3 Suggested slide outline (about 12 slides)

1. Title, component, supervisor.
2. Problem: midday surplus and evening peak; 2025 curtailment events; your midday-dip analysis.
3. Gap and objectives SO1-SO6 (one line each, with status).
4. Method pipeline (Figure 1) and the five knowledge pillars.
5. Data: PUCSL vs CEB reconciliation; the rooftop-solar discovery.
6. NLmin from the LTGEP and the scenario-year table (risk emerges by 2028).
7. The LP scheduler: constraints, epigraph objective, audit; technologies.
8. Results on the risk days: table and Figure 2b (9 June).
9. Trade-off (a step, not a curve) and sensitivity (never outranked in 16/16 settings).
10. Requirements traceability and completion against the WBS.
11. Risks and corrective actions.
12. Commercialisation and next steps.

Appendix A. Glossary

Term	Meaning
Curtailment	Deliberately reducing renewable output that was available, because the grid cannot use it safely
Synchronous generator	A spinning machine (coal, oil, hydro) locked to grid frequency; provides inertia
Inertia	Stored rotating energy that slows frequency changes after a fault; measured in MW·s
SNSP	System non-synchronous penetration: share of demand met by inverter-based sources
Must-take	Generation that cannot be turned down by the operator (run-of-river mini hydro, biomass)
Embedded generation	Small generators connected to the distribution network, not seen in transmission dispatch
Residual / net load NL(t)	Demand minus variable renewables and must-take; served by dispatchable plants
NLmin	Minimum secure level of synchronous generation
C(t)	Inferred curtailment risk (MW); a scenario signal, not a measurement
Linear program (LP)	Optimisation with a linear objective and linear constraints; solved to a proven optimum
Epigraph form	Replacing max(·) by a variable bounded below by each argument, inside a minimisation
HiGHS	Open-source LP solver used through SciPy
GMM	Gaussian Mixture Model: a probability model made of several bell-shaped components
BIC	Bayesian Information Criterion: chooses model size by trading fit against complexity
Persistence forecast	Tomorrow equals today at the same hour; the standard naive benchmark
MAE / RMSE	Mean absolute error / root mean squared error of a forecast
Skill	$1 - \mathrm{MAE}_{\mathrm{model}} / \mathrm{MAE}_{\mathrm{persistence}}$; above 0 means better than persistence
Precision / recall	Share of flagged hours that are right / share of real events that are found
Chronological hold-out	Train on earlier months, test on later months the model never saw

Appendix B. How to run everything

From the project folder in PowerShell (raw data must already be in data/raw):

```
.\.venv\Scripts\python.exe -m pytest -q
.\.venv\Scripts\python.exe scripts\build_hourly_net_demand.py --input
data\processed\grid\pucsl_2024_consolidated_dispatch_15min.csv --year 2024
.\.venv\Scripts\python.exe scripts\build_grid_scenario.py
.\.venv\Scripts\python.exe scripts\derive_nlmin.py
.\.venv\Scripts\python.exe scripts\forecast_grid.py
.\.venv\Scripts\python.exe scripts\run_experiments.py --quick --workers 4
.\.venv\Scripts\python.exe scripts\run_experiments.py --workers 6
.\.venv\Scripts\python.exe scripts\benchmark_scaling.py
.\.venv\Scripts\streamlit.exe run app\streamlit_app.py (prototype at http://localhost:8501)
.\.venv\Scripts\python.exe scripts\run_experiments.py --workers 6 --acn-json
data\raw\charging_sessions\acn.json
```

Output	Where
Boundary-corrected hourly grid	data/processed/grid/grid_base_2024_hourly.csv
Grid diagnostics and scenario years	results/grid_scenario/grid_base_2024_diagnostics.json
NLmin derivation	results/grid_scenario/nlmin_derivation.json
Risk-day comparison; hourly EV loads	results/experiments/so5_risk_days_comparison.csv; so5_hourly_ev_load_mw.csv
Forecasts and accuracy	data/processed/grid/grid_forecast_2024_hourly.csv; results/forecast/
Strategy comparison	results/experiments/so5_comparison.csv

Output	Where
Trade-off and sensitivity	results/experiments/so5_tradeoff.csv; so6_sensitivity.csv
Run record (config, git version, days)	results/experiments/run_summary.json
Runtime benchmark (NFR1, NFR2)	results/experiments/scaling_benchmark.csv
Figures	results/experiments/figures/

Appendix C. Parameters (config/scenarios.json)

Parameter	Central value	Sweep	Source
NLmin	520 MW	240, 380, 520, 650, 780 MW	derived (scripts/derive_nlmin.py): 2 coal units x 150 MW minimum (LTGEP Table 2.7) + 217 MW major hydro kept online (observed 2024 practice); pending CEB review
SNSP limit	0.75	off, 0.5, 0.65, 0.75	CEB LTGEP 2025-2044 planning limit, pp. 7-7, 7-13, 10-14; 2024 operation never exceeded 46%
Scenario year	2028	2026, 2027, 2028, 2030	LTGEP base case; multipliers derived by build_grid_scenario.ltgep_scenario
Sessions per day	10,000	-	provisional: national public sessions/day; replace with Component 1 fleet scenario
Station capacity	50 kW, 8 sessions/day	-	provisional: 50 kW site connection shared by its charge points
Declared-departure share	0.8	0.5, 0.8, 1.0	provisional
Objective weights	curtailment 1, peak 1.0, delay 0.01	peak 0-16; delay 0-0.1	swept (no single 'right' weighting)

Appendix D. Likely viva questions, with answers

Q. Why don't you use measured curtailment data?

A. None is published for Sri Lanka. That absence is the research gap: the signal is inferred from the constraints that cause curtailment, following Denholm & Margolis (2007).

Q. Why did you model solar instead of using the metered values?

A. Metered solar is recorded after curtailment, so it hides exactly what we want to measure. Available solar is modelled from NASA irradiance and scaled to CEB's official monthly energy.

Q. How do you know your demand series is right?

A. It is reconciled to the CEB Data Book: corrected demand matches CEB net generation within 0.1%, and coal matches within 0.2%.

Q. Why not add minimum generation, reserve and inertia to get NLmin, as in your proposal?

A. That double-counts the same running units, and inertia is measured in MW·s, not MW. The floor plus an SNSP limit is the form used by island operators such as EirGrid, and by CEB's own plan.

Q. Where does 520 MW come from?

A. Two Lakvijaya units at their 150 MW minimum operating level (LTGEP Table 2.7) plus about 217 MW of major hydro, which is what the operator kept online in the 167 most stressed hours of 2024. It passes the LTGEP 5% reserve and 25% single-unit rules, and the real system never ran below it in 2024. It is swept from 240 to 780 MW, and I have asked CEB to review it.

Q. Why scenario year 2028?

A. With the LTGEP base-case solar build-out and demand forecast, risk is negligible until 2026 and becomes material in 2028. 2026, 2027 and 2030 are run as a sensitivity, so the conclusion does not depend on one year.

Q. Why linear programming and not machine learning for scheduling?

A. The problem is linear, the LP gives a proven optimum, needs no training data and is reproducible. ML is used where it fits: forecasting tomorrow's grid conditions.

Q. What models did you train?

A. Day-ahead forecasting models (gradient boosting for demand, ridge for wind and must-take), validated on a chronological hold-out against persistence. The session GMM will be trained on ACN-Data. The LP and C(t) are solved or computed, not trained.

Q. On most days your scheduler absorbs the same as solar-following. So what is your contribution?

A. When the risk sits on the sunniest hours both target the same hours, so absorption is equal; but the curtailment-aware schedule gets it with about a third of the driver delay and twice the evening-peak reduction. When wind or the demand shape moves the risk away from the solar peak, as on 9 June 2024 in the monsoon, solar-following misses it and curtailment-aware does not. Reporting service cost together with grid benefit is the point of the evaluation.

Q. How do you guarantee every driver gets their energy?

A. Full energy is a hard LP constraint; requests are capped at what the charger can deliver; an infeasible day is reported, never relaxed; and an independent audit re-checks every schedule.

Q. Is ACN-Data appropriate for Sri Lanka?

A. Not directly: it is US workplace charging. Its parameters are treated as scenario variables and tested in sensitivity analysis; local operator data would replace it if obtained.

Q. How robust are your conclusions?

A. The analysis is repeated in 16 settings across NLmin, SNSP, scenario year and declared-departure share. Curtailment-aware is never outranked in any of them; where the exact order changes, it is because curtailment-aware moves from tied to ahead. The most sensitive input is how many drivers declare a departure time: below 80% many days become infeasible at the station.

Q. What is novel?

A. Scheduling public EV charging against an inferred, constraint-derived curtailment signal in an island system with no curtailment record, plus a reconciled Sri Lankan grid dataset and a measured forecast-value result.